Certificate of Advanced Studies (CAS) in Advanced Machine Learning

Final Project Report

Sensor Fusion for 3D Object Detection in Autonomous Vehicles

Evaluation on the nuScenes Benchmark Dataset

submitted to

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1 Introduction

3D object detection is a key component of autonomous driving systems, enabling vehicles to detect and localize surrounding objects such as cars, pedestrians, and cyclists. Accurate environment perception is essential for safe navigation and decision-making.

Modern autonomous vehicles rely on multiple sensing modalities, including cameras, LiDAR, and radar. The nuScenes dataset, introduced by Caesar et al. [1], is one of the most widely used benchmarks for autonomous driving research. It provides synchronized camera, LiDAR, and radar data together with detailed 3D object annotations.

The objective of this project is to investigate the impact of sensor fusion on 3D object detection performance using the nuScenes dataset. An exploratory analysis of the dataset is first conducted to characterize object observability and sensor support. The study then compares the LiDAR-only CenterPoint [2] with the camera–LiDAR fusion BEVFusion model [3]. The evaluation focuses on the influence of sensing modality and training dataset size using the official nuScenes detection metrics.

2 The data

2.1 nuScenes Dataset

The nuScenes dataset is a large-scale autonomous driving dataset designed for 3D perception tasks. It contains urban driving scenes collected in Boston and Singapore under diverse traffic, weather and lighting conditions.

As shown in Figure 1, nuScenes provides synchronized multimodal sensor data with full 360° coverage around the ego vehicle. The sensor suite consists of six RGB cameras, one LiDAR sensor, and five radar sensors, providing complementary visual, geometric, and motion information about the environment.

The dataset is organized as a relational database linking scenes, sensor measurements, and object annotations (Figure 2). Table 1 summarizes the trainval split used in this study, while Table 2 presents the 23 object categories defined in the nuScenes taxonomy.

The diversity of sensing modalities, environments and annotated objects makes nuScenes a widely used benchmark for evaluating 3D object detection and sensor fusion methods.

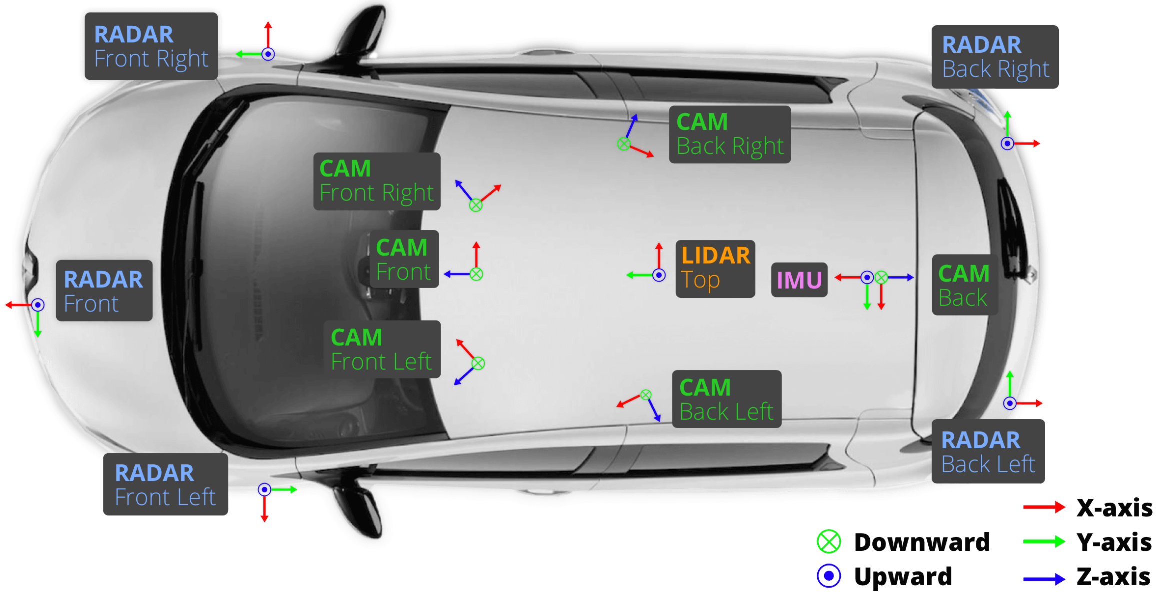

Figure 1 - Sensor configuration of the nuScenes dataset. [1].

Dataset Component	Count
Scenes	850
Samples	34,149
3D Annotations	1,166,187
Object Instances	64,386
Categories	23
Sensors	12
Sensor Measurements	2,631,083

Table 1 - Summary of the nuScenes trainval dataset.

[1] Source: https://www.nuscenes.org/static/media/data.9ef46c59.png

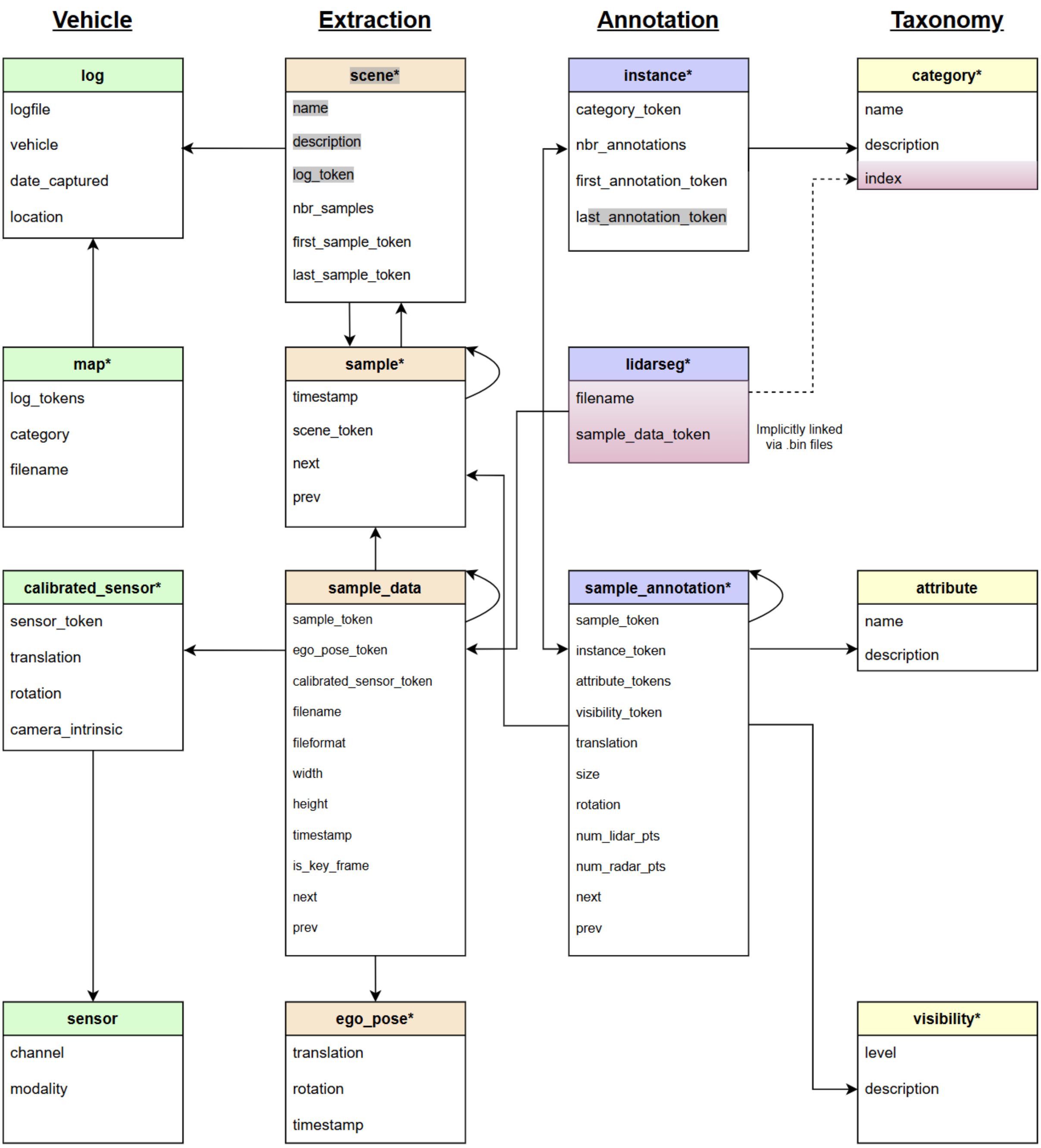

Figure 2 - Relational schema linking scenes, sensor measurements, ego-vehicle poses, and object annotations in the nuScenes dataset [2]

[2] https://www.nuscenes.org/public/images/nuscenes-schema.svg

Group	Categories
Human pedestrian	adult, child, construction worker, personal mobility, police officer, stroller, wheelchair
Movable object	barrier, debris, pushable_pullable, trafficcone
Vehicle	bicycle, bus_bendy, bus_rigid, car, construction, emergency_ambulance, emergency_police, motorcycle, trailer, truck
Static object	bicycle_rack
Animal	animal

Table 2 – The 23 object categories defined in the nuScenes taxonomy.

3 Exploratory Data Analysis

3.1 EDA Objectives and Methodology

The objective of the exploratory data analysis (EDA) was to characterize the nuScenes dataset and assess how camera, LiDAR, and radar observations vary with object characteristics and observation conditions. The analysis focused on object geometry, visibility, spatial distribution, and sensor support to better understand the factors that influence 3D object detection.

3.2 Feature engineering

The exploratory analysis required additional features derived from the nuScenes annotation and sensor data. As summarized in Table 3, these features were designed to characterize object distance, geometry, and sensor observability.

The distance to the ego vehicle was computed in the horizontal plane from the object and ego-vehicle positions. Geometric features were derived from the 3D bounding-box dimensions, including box volume and length-to-width aspect ratio.

Sensor-support features were computed separately for LiDAR, radar and camera data. LiDAR and radar support were based on the number of sensor points associated with each annotation. For LiDAR, annotations were grouped into no support, low support and usable support using thresholds of 0, 1–4 and at least 5 points respectively. For radar, the corresponding thresholds were 0, 1–2 and at least 3 points.

Camera visibility was estimated by projecting each 3D bounding box into the available camera views. For each annotation, the number of visible cameras, the projected box area and the best camera view were computed. These features were then used to compare the observability of objects across sensing modalities.

Feature group	Feature / definition	Purpose

Distance	distance_ego_2d = 2D distance between object center and ego vehicle	Analyze object range
Geometry	Box volume = width × length × height	Compare object size
Geometry	Aspect ratio = length / width	Characterize object shape
LiDAR support	No: 0 points; Low: 1–4 points; Usable: ≥5 points	Estimate LiDAR observability
Radar support	No: 0 points; Low: 1–2 points; Usable: ≥3 points	Estimate radar observability
Camera visibility	Number of visible cameras	Estimate camera coverage
Camera visibility	Maximum / mean projected box area in pixels	Estimate image visibility and apparent object size
Camera visibility	Best camera view	Identify the camera with the largest object projection

Table 3 - Engineered features used for exploratory data analysis

3.3 Detection Classes

Figure 3 and Figure 4 show a highly imbalanced class distribution in the nuScenes dataset. Vehicle and pedestrian categories dominate the annotations, while classes such as emergency vehicles, wheelchairs, and personal mobility devices are rare.

Cars alone account for more than 40% of all annotations, followed by adult pedestrians, barriers, and traffic cones. Together, these categories represent most objects in the dataset.

This imbalance reflects real-world urban traffic and results in substantial differences in class representation across the dataset.

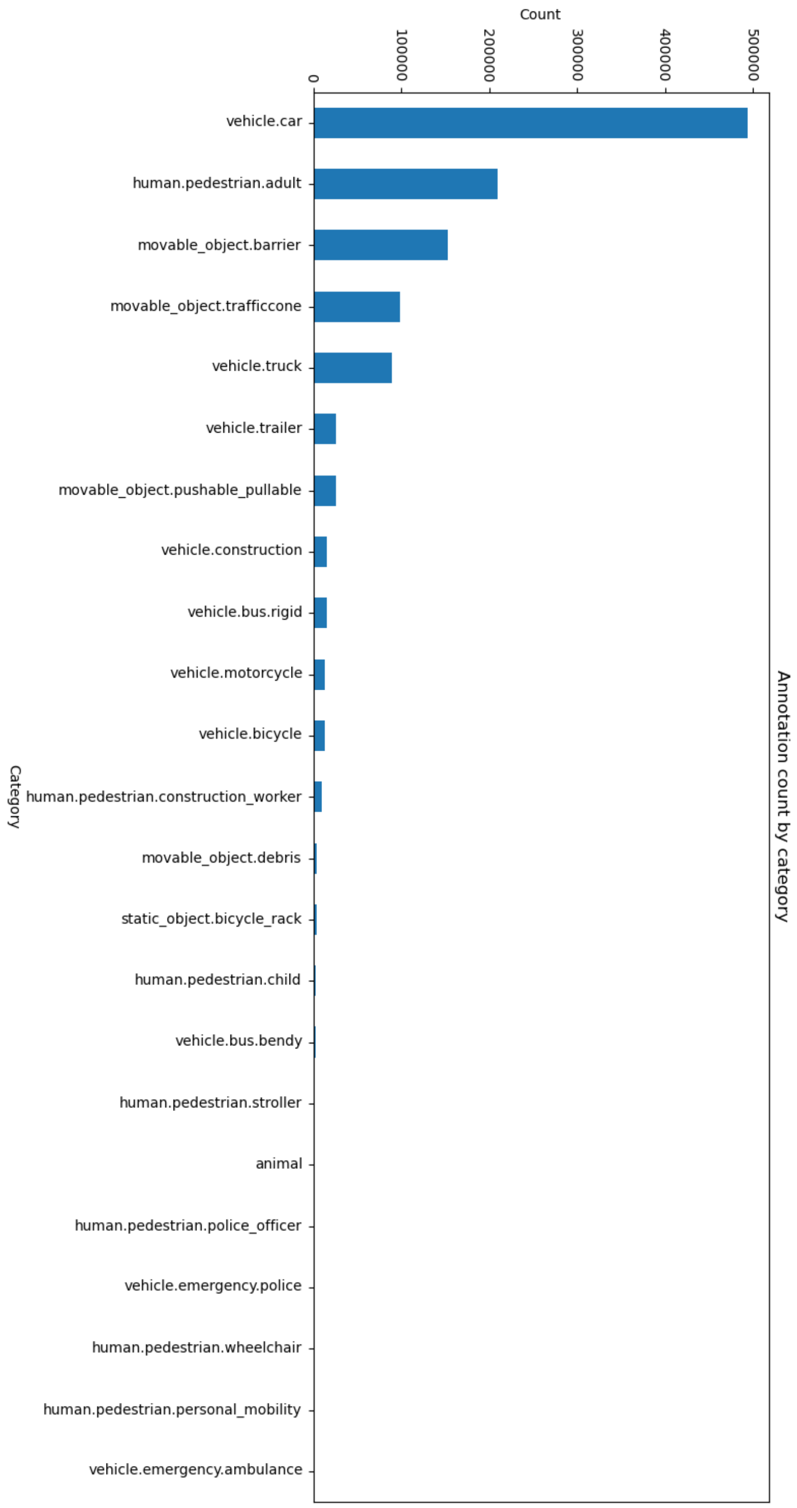

Figure 3 - Annotation count by category

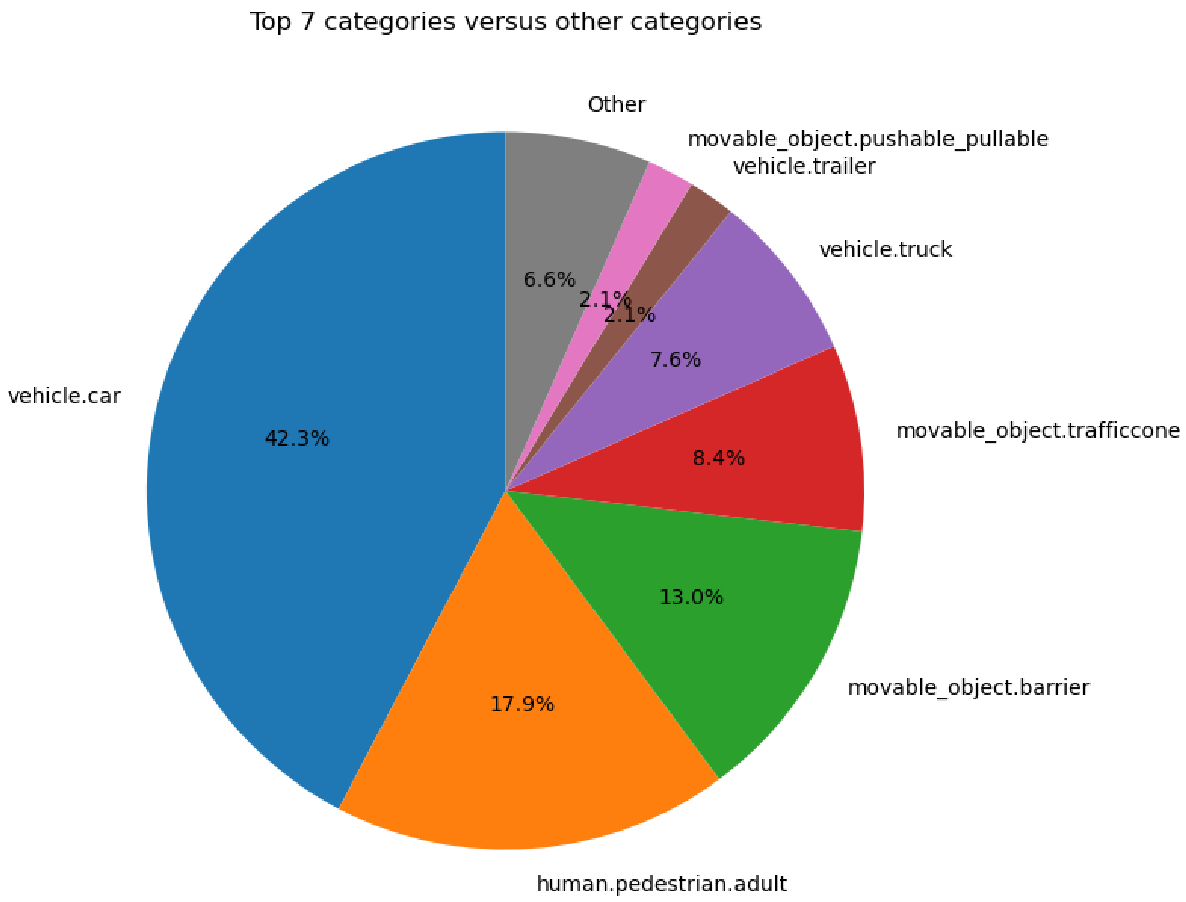

Figure 4 - Top 7 categories versus remaining categories

3.4 Visibility Analysis

Object visibility is an important factor for 3D object detection because partially occluded objects are generally more difficult to detect than fully visible objects. In nuScenes, each annotation is assigned to one of four visibility levels corresponding to the estimated fraction of the object that is visible: 0–40%, 40–60%, 60–80% and 80–100%.

Figure 5 shows the distribution of visibility levels across object categories. Most categories are dominated by highly visible objects (80–100%), indicating that the majority of annotated objects are only weakly occluded. However, noticeable differences exist between categories. Pedestrians, bicycles and motorcycles exhibit a larger proportion of partially visible instances, while several vehicle categories contain a higher proportion of highly visible objects.

These results show that visibility conditions vary across object categories, with pedestrians, bicycles, and motorcycles generally exhibiting higher levels of occlusion than larger vehicle classes.

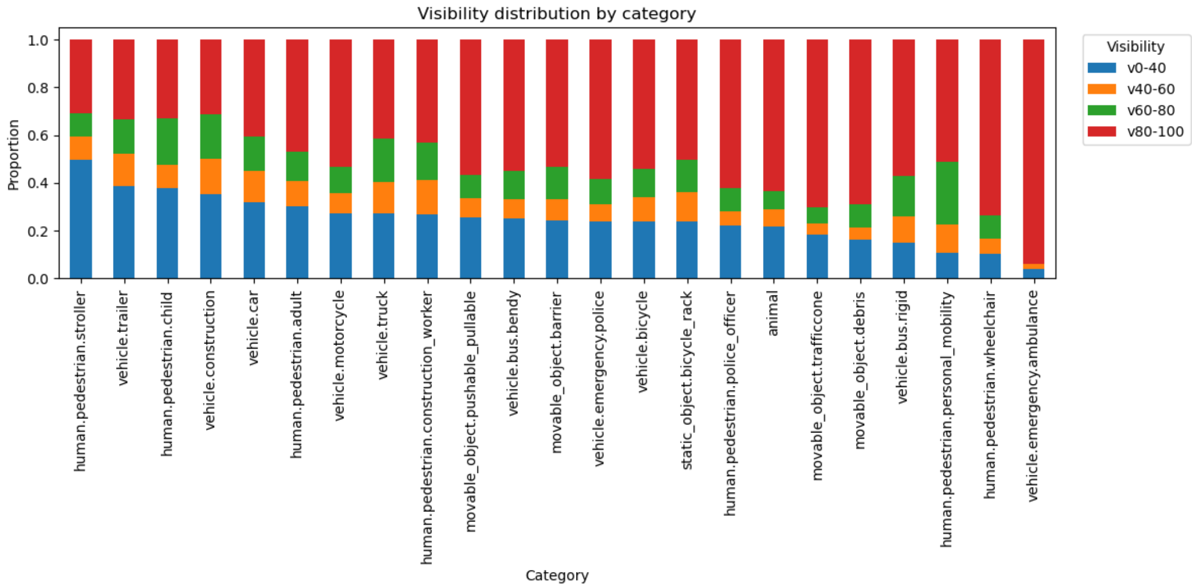

Figure 5- Visibility distribution by object category in the nuScenes dataset. Visibility is defined as the estimated fraction of the object visible in the scene.

3.5 Object Characteristics

Object geometry strongly influences 3D object detection, as smaller objects generate fewer image pixels and LiDAR returns.

Figure 6 shows that object volumes span several orders of magnitude. Large vehicles (trailers, buses, trucks, construction vehicles) occupy the largest volumes, while traffic cones, debris, and pedestrian-related categories are among the smallest. Some categories, particularly construction vehicles and trailers, also exhibit substantial size variability.

Figure 7 compares median object dimensions. Large vehicle classes are considerably longer and wider than pedestrians and small roadside objects, while bicycles and motorcycles occupy an intermediate range.

Overall, the analysis highlights substantial geometric diversity across object categories, with object volumes spanning several orders of magnitude.

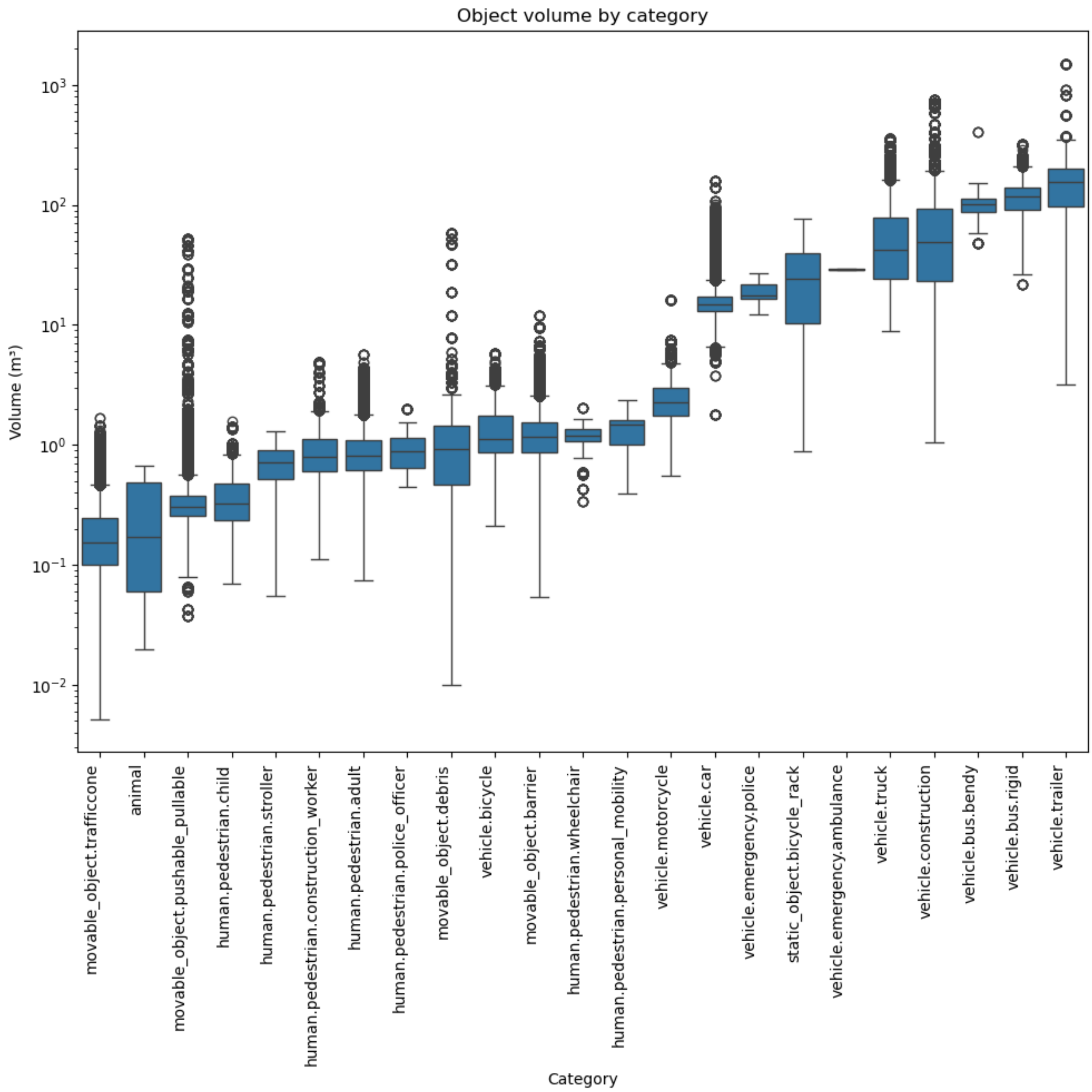

Figure 6 - Distribution of object volume by category.

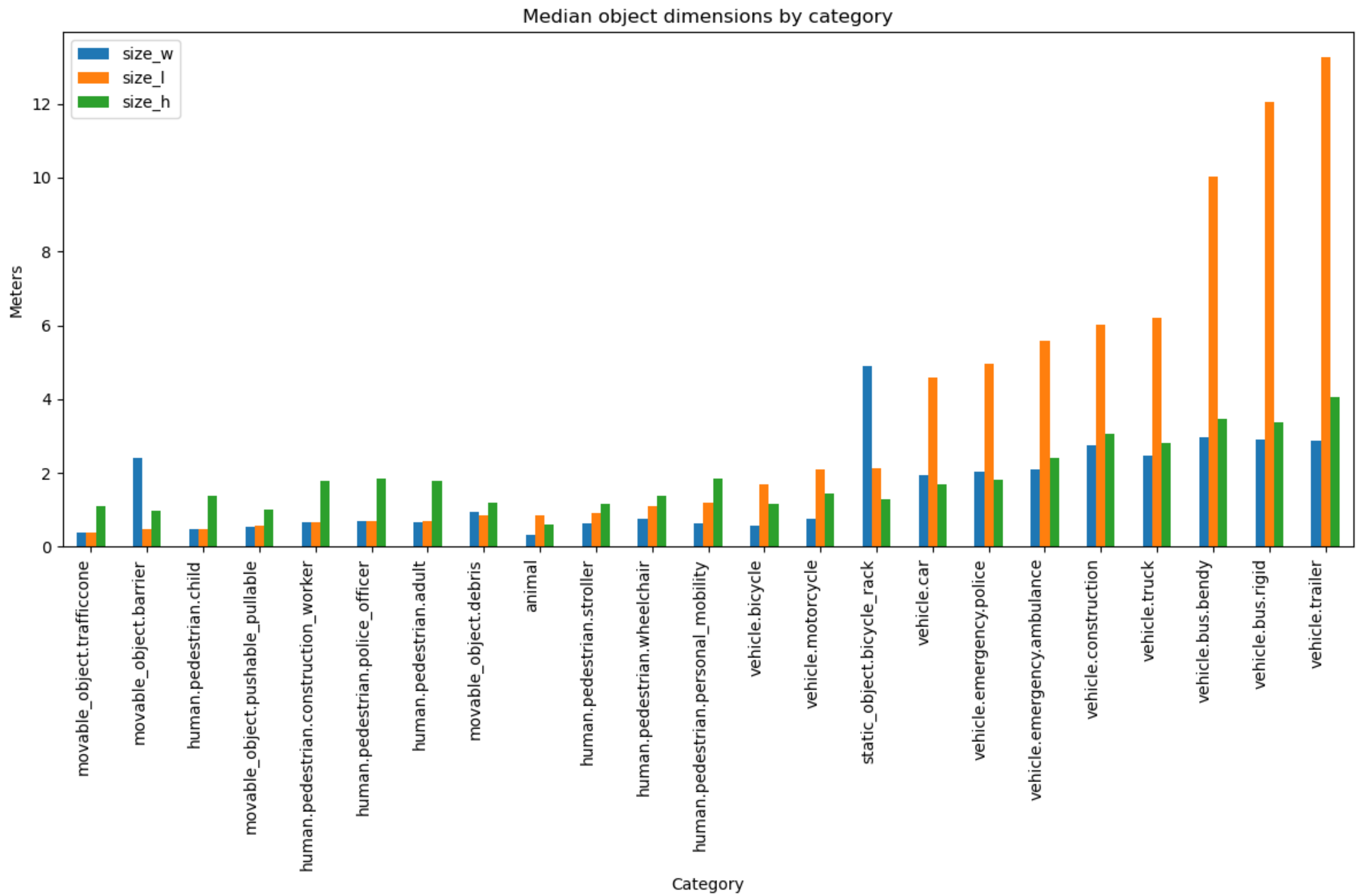

Figure 7 - Median object dimensions by category.

3.6 Spatial Distribution

Object distance directly affects sensor observability and detection performance. Nearby objects provide stronger visual and geometric evidence, while distant objects generate fewer sensor measurements.

Figure 8 shows that most annotations are located within 50 m of the ego vehicle, with a peak between 20 and 35 m. The number of observations decreases progressively at greater distances, forming a long-tailed distribution extending beyond 300 m.

Overall, the nuScenes dataset is dominated by short- and medium-range observations, with the number of annotated objects decreasing progressively at greater distances.

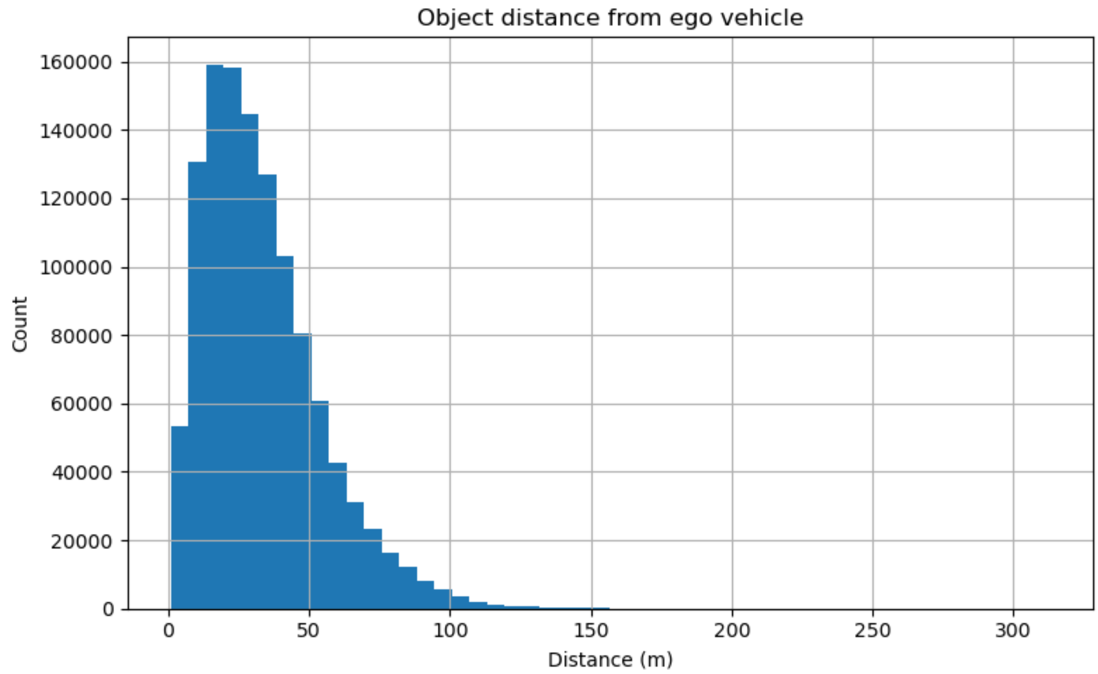

Figure 8 - Distribution of object distance from the ego vehicle.

3.7 Sensor Support Analysis

3.7.1 Support by distance

Figure 9 shows that camera and LiDAR provide strong usable support at short distances, with more than 95% of objects receiving usable measurements within 10 m of the ego vehicle. However, support decreases progressively with distance and drops substantially beyond 50 m. Radar provides considerably lower usable support across all distance ranges.

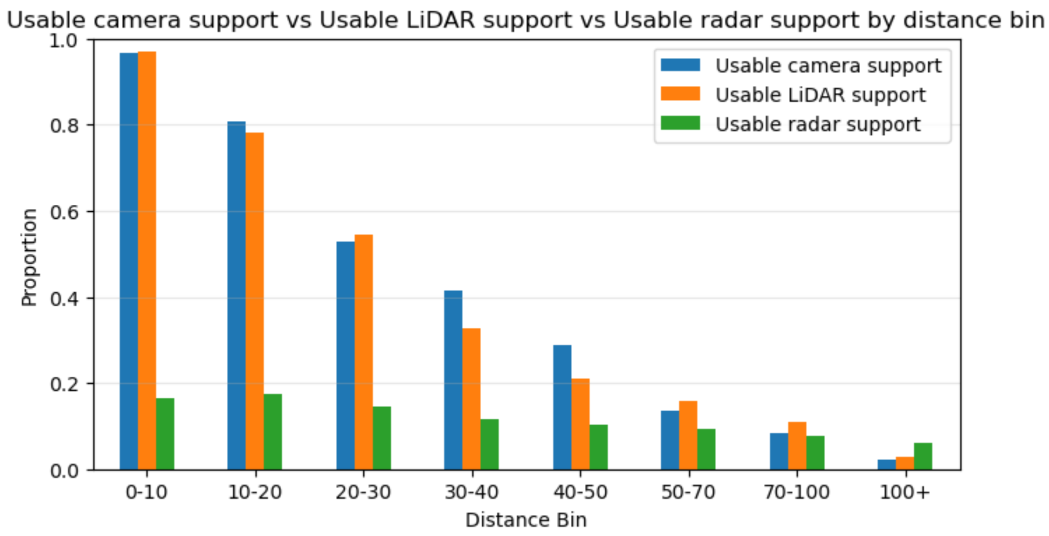

Figure 9 - Proportion of objects receiving usable sensor support as a function of distance.

Figure 10 shows that the proportion of low-support objects increases with distance for camera and LiDAR at short and intermediate ranges before decreasing at longer distances. In contrast, radar remains relatively stable across distance ranges.

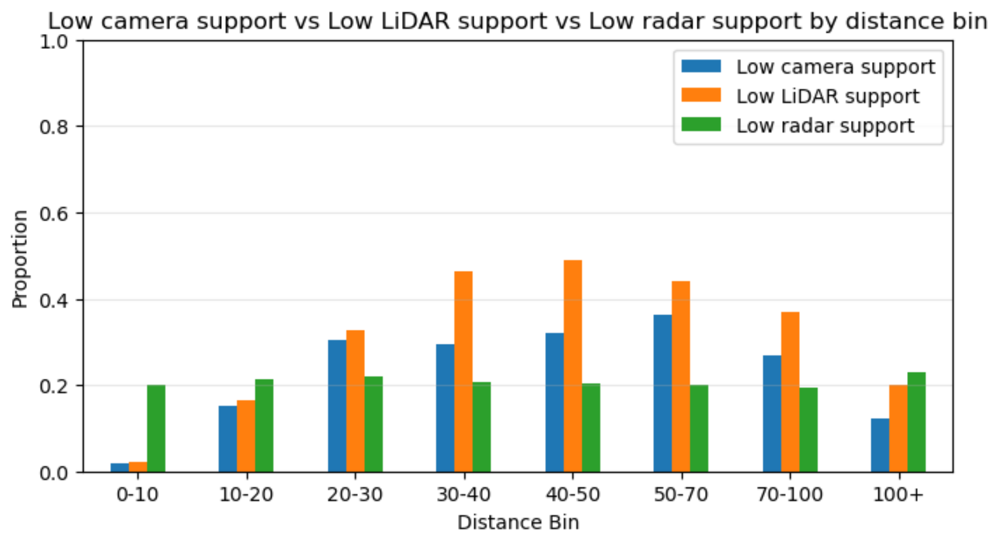

Figure 10 - Proportion of objects receiving low sensor support as a function of distance.

Figure 11 shows that no-support rates increase steadily with distance for all modalities and become dominant beyond approximately 70 m for camera and LiDAR. Radar exhibits the highest no-support rate throughout the entire distance range.

Overall, object observability decreases with distance for all three sensing modalities. Camera and LiDAR maintain high levels of usable support at short and medium ranges, whereas radar provides substantially fewer object-level observations across all distance bins.

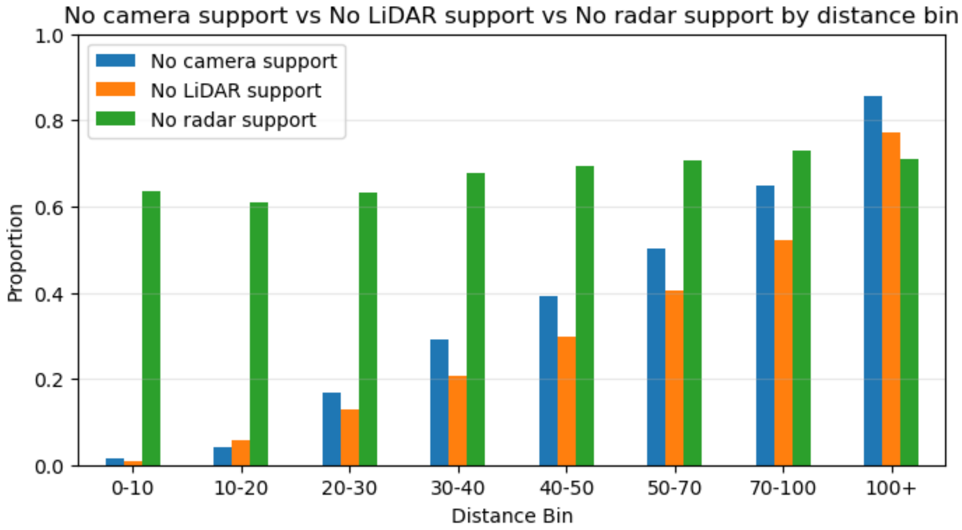

Figure 11 - Proportion of objects receiving no sensor support as a function of distance.

3.7.2 Support by visibility

Object visibility affects sensor observability, as occluded objects provide less information than fully visible ones. To evaluate this effect, sensor support was analyzed across the four nuScenes visibility levels.

Figure 12 shows that usable support increases with visibility for all modalities. The effect is particularly pronounced for LiDAR, where usable support increases from approximately 20% for heavily occluded objects to more than 60% for highly visible objects, indicating that LiDAR observability is strongly influenced by object visibility. Camera support also improves with visibility, while radar remains limited across all visibility levels.

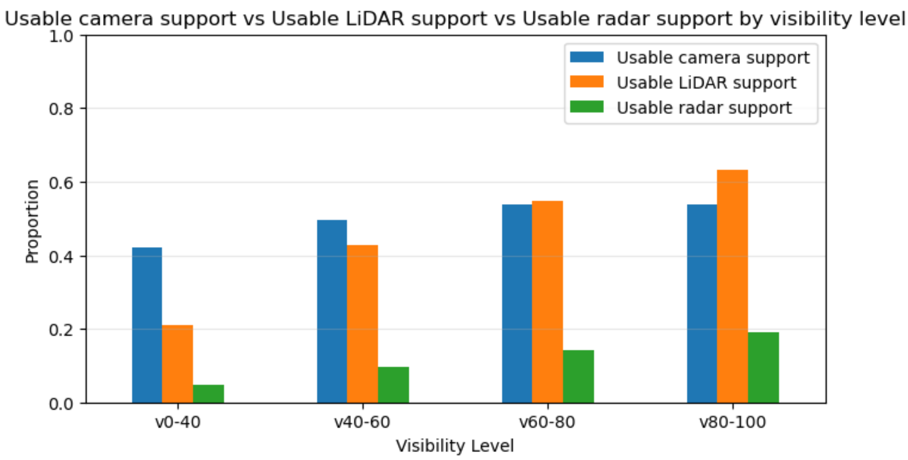

Figure 12 - Proportion of objects receiving usable sensor support as a function of visibility level.

Figure 13 shows that the proportion of low-support LiDAR observations decreases moderately with increasing visibility, from approximately 40% for heavily occluded objects to about 27% for highly visible objects. In contrast, camera low-support rates decrease only slightly, while radar remains relatively stable across visibility levels.

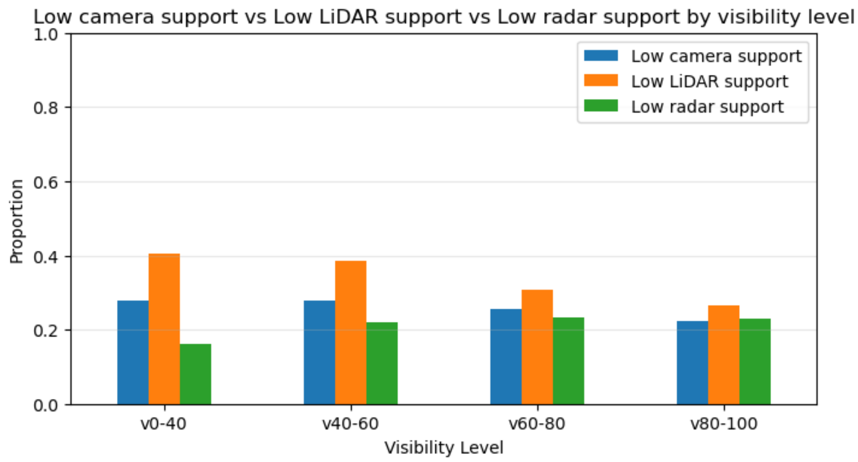

Figure 13 - Proportion of objects receiving low sensor support as a function of visibility level.

Figure 14 shows that LiDAR no-support rates decrease markedly with increasing visibility, whereas camera no-support rates remain relatively stable across visibility levels. Radar no-support rates also decrease with visibility but remain substantially higher than those of camera and LiDAR, even for highly visible objects.

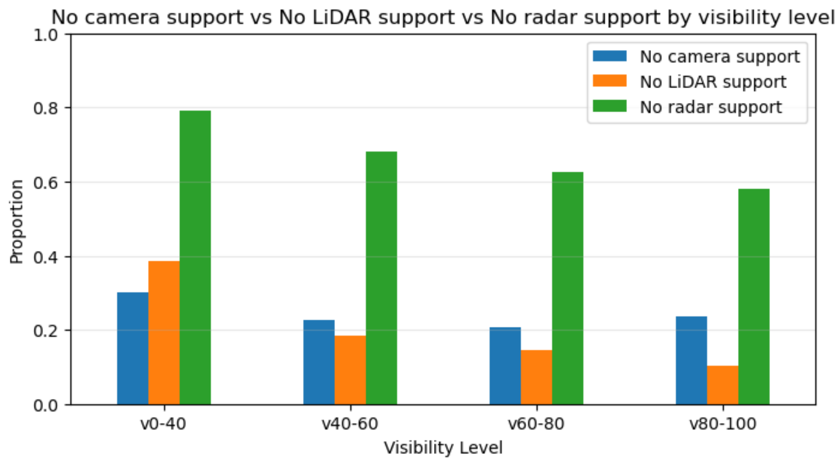

Figure 14 - Proportion of objects receiving no sensor support as a function of visibility level.Support by category

Overall, visibility has a strong influence on sensor observability, particularly for LiDAR. Improved visibility substantially increases usable LiDAR support while reducing low-support and no-support observations. For cameras, visibility is primarily associated with an increase in usable support, whereas low-support and no-support rates vary more moderately across visibility levels. In contrast, radar support remains comparatively limited across all visibility levels.

3.7.3 Support by Category

Sensor observability depends not only on distance and visibility but also on object category, as object size and geometry influence the amount of sensor information available.

[Figure 15](#) shows that usable camera and LiDAR support varies considerably across object categories. Large vehicle classes, such as buses, trailers, trucks, and construction vehicles, generally receive the strongest support from both modalities. In contrast, traffic cones and several pedestrian-related and movable-object categories receive usable support less frequently. Radar provides substantially less usable support overall and contributes primarily for larger vehicle classes.

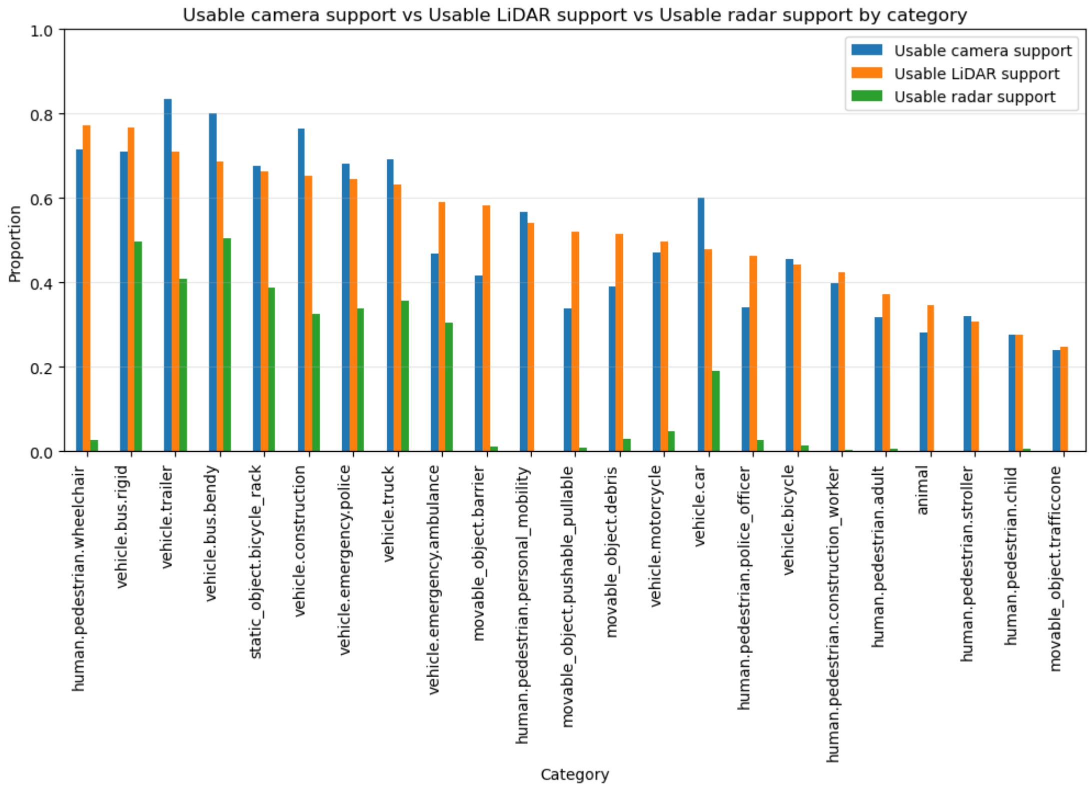

Figure 15 - Proportion of objects receiving usable sensor support by category.

Figure 16 shows that pedestrian-related categories, traffic cones, and several small objects are more frequently associated with low-support camera and LiDAR observations than most large vehicle classes.

Figure 17 shows that radar exhibits the highest no-support rates across most categories. Camera and LiDAR no-support rates are generally lower, although non-zero no-support rates are observed across all three modalities.

Overall, sensor observability varies considerably across object categories. Large vehicle classes generally receive the strongest camera and LiDAR support, whereas traffic cones and several pedestrian-related and movable-object categories are more frequently associated with low-support or no-support observations. Radar support remains comparatively limited across most categories.

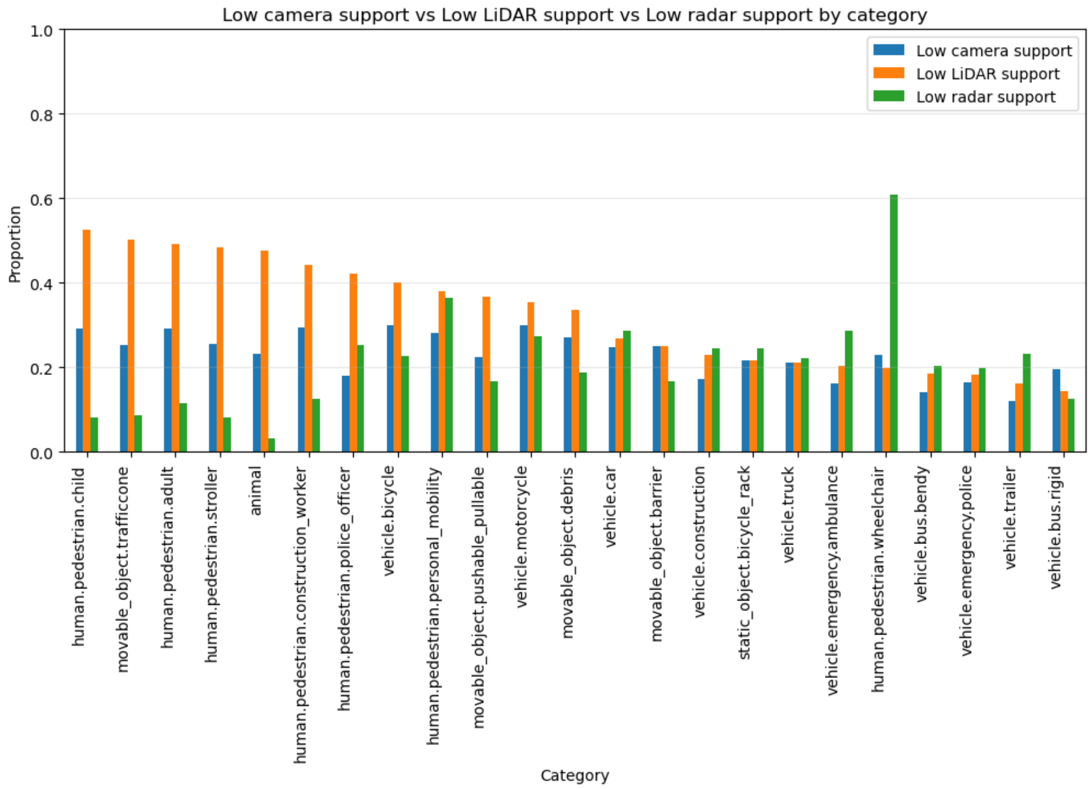

Figure 16 - Proportion of objects receiving low sensor support by category.

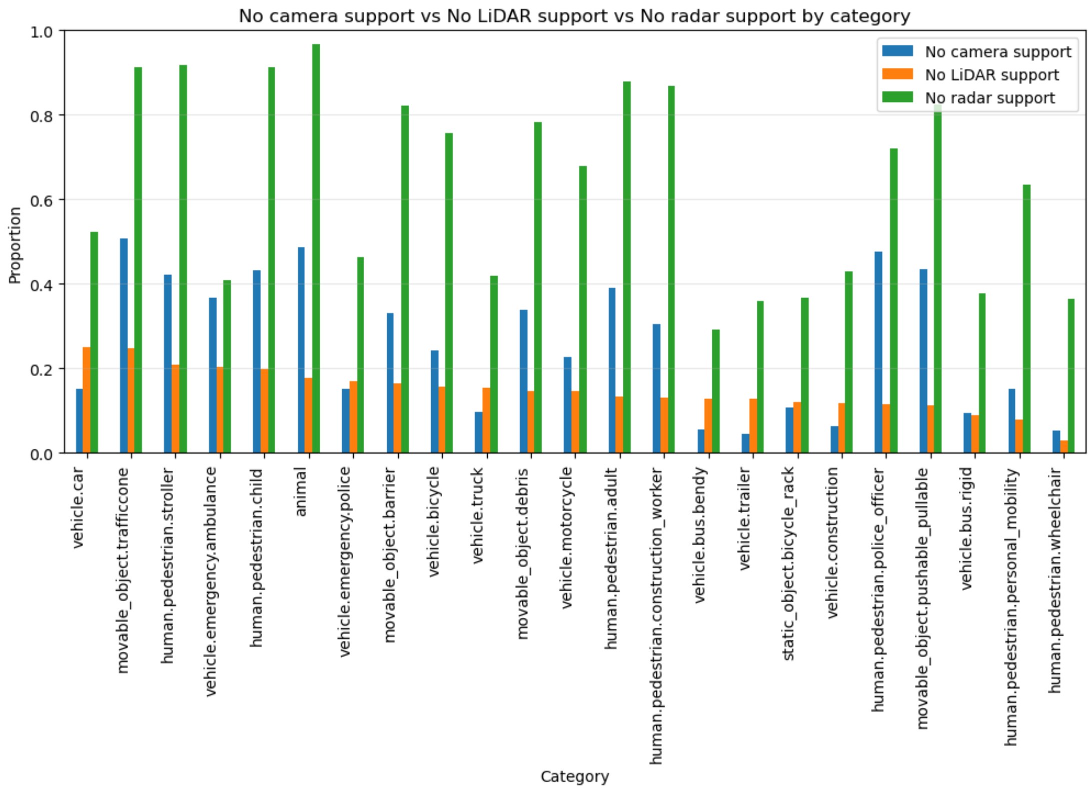

Figure 17 - Proportion of objects receiving no sensor support by category.

3.7.4 Camera–LiDAR Complementarity

The previous analyses showed that camera and LiDAR provide substantially stronger object-level support than radar across distance ranges, visibility conditions and object categories. To further investigate their complementarity, two representative support regimes were defined:

- ***Camera strong with LiDAR support*****:** objects receiving usable camera support together with non-zero LiDAR support.
- **Camera not good and LiDAR low:** objects receiving weak camera support together with low LiDAR support.

These regimes were analyzed to quantify how frequently camera and LiDAR provide complementary information under different observation conditions.

Overall, 45.5% of all annotations belong to the Camera strong with LiDAR support regime, while 24.75% belong to the Camera not good and LiDAR low regime. Together, these two mutually exclusive regimes account for approximately 70.3% of all annotated objects.

Figure 18 shows the prevalence of the two regimes across distance ranges. The proportion of objects receiving strong camera support together with LiDAR support decreases steadily with increasing distance, from more than 95% at short range to nearly zero beyond 100 meters. Conversely, the proportion of objects receiving weak joint support increases with distance and becomes dominant beyond approximately 40 m, indicating that the simultaneous availability of useful camera and LiDAR observations deteriorates rapidly at longer ranges.

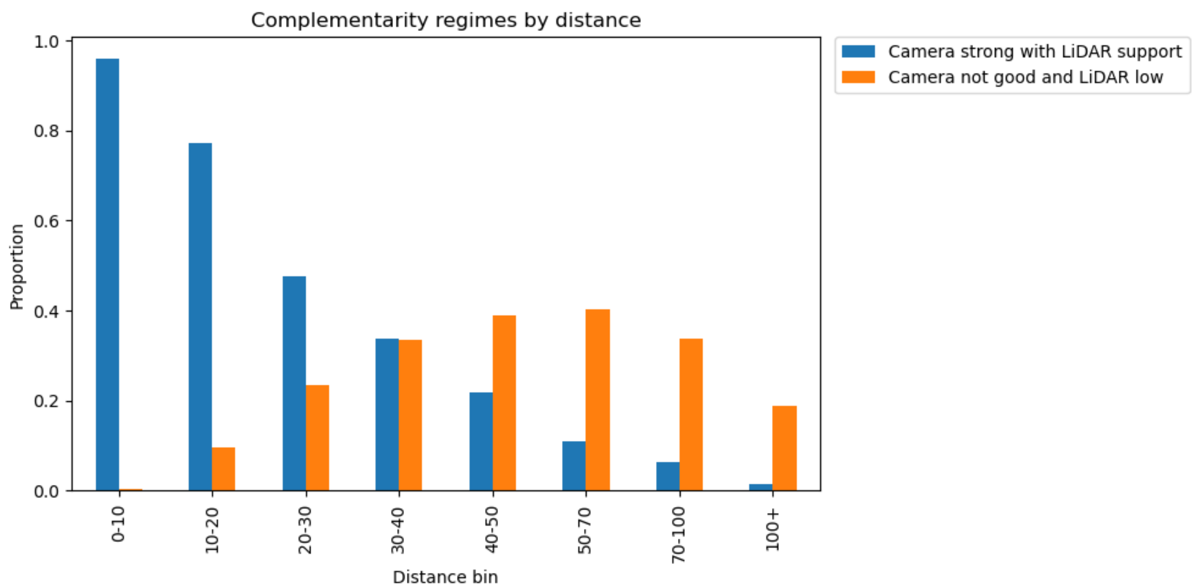

Figure 18 - - Camera–LiDAR complementarity regimes by distance.

Figure 19 presents the same analysis across visibility levels. The prevalence of the Camera strong with LiDAR support regime increases consistently with visibility, while the Camera not good and LiDAR low regime decreases slightly.

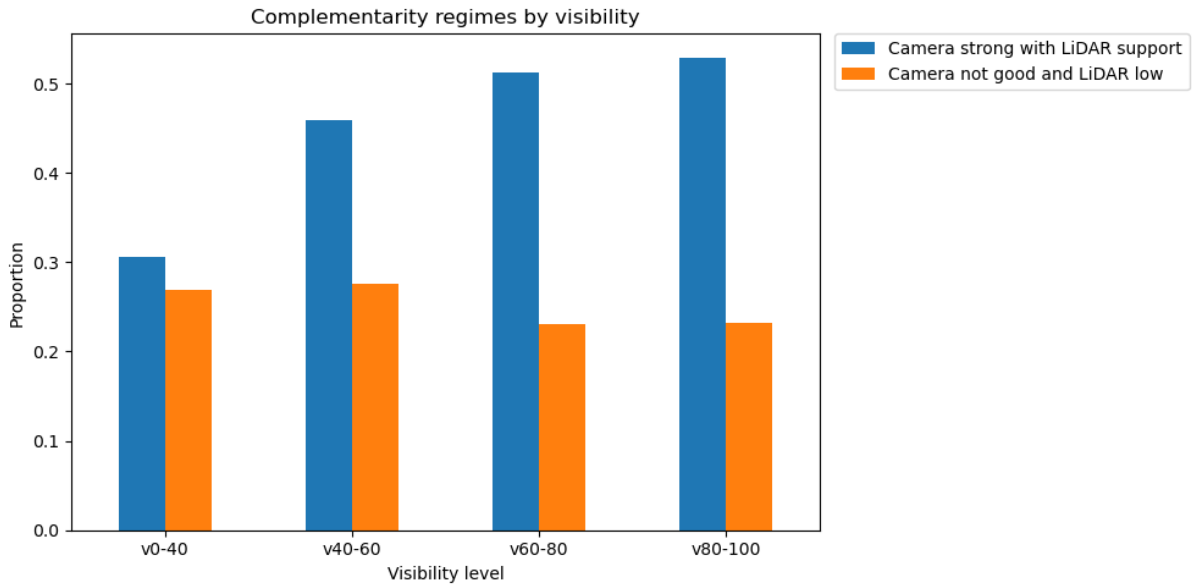

Figure 19 - Camera–LiDAR complementarity regimes by visibility level.

These results show that camera–LiDAR support regimes vary systematically with distance and visibility. Strong joint support is most prevalent for nearby and highly visible objects, whereas weaker joint support is more frequently observed for distant and partially occluded objects.

3.7.5 Category level complementarity

The complementarity analysis was extended to object categories to identify which classes benefit most from combined camera and LiDAR observations.

Figure 20 shows the distribution of Camera–LiDAR support regimes across categories. Large vehicle classes, such as buses, trailers, and trucks, exhibit the highest proportion of the Camera strong with LiDAR support regime. In contrast, pedestrian-related categories, traffic cones, and several movable-object classes show lower levels of joint support and a higher prevalence of the Camera not good and LiDAR low regime.

These results indicate that Camera–LiDAR support varies substantially across object categories, with large vehicle classes generally exhibiting stronger joint support than pedestrian-related and other small-object categories.

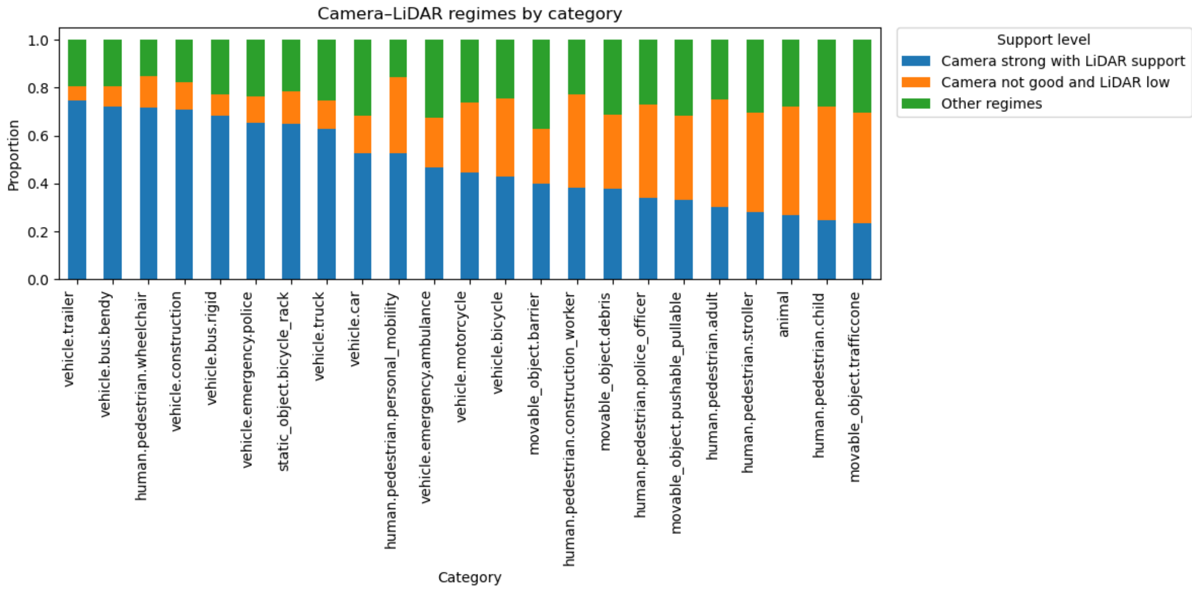

Figure 20 - Camera–LiDAR complementarity regimes by category.

3.8 Key Findings

The exploratory data analysis revealed substantial variability in object observability across the nuScenes dataset. This variability arises from differences in category frequency, object geometry, visibility conditions, and distance from the ego vehicle.

The analysis showed that sensor support varies systematically with distance and visibility. For camera and LiDAR, usable support decreases with distance and increases with visibility. Low-support observations become more prevalent at intermediate distances and lower visibility levels, whereas no-support observations become increasingly common for distant and poorly visible objects. Most objects are located within 50 m of the ego vehicle, where the availability of usable sensor observations is highest.

Camera and LiDAR consistently provide stronger object-level support than radar across distances, visibility levels, and object categories. The complementarity analysis showed that approximately 45.5% of annotations belong to the Camera strong with LiDAR support regime, indicating that both modalities frequently provide usable support simultaneously. In contrast, 24.75% of annotations belong to the Camera not good and LiDAR low regime, where both modalities provide only limited support. The prevalence of these regimes varies considerably with distance, visibility, and object category.

Finally, the combination of geometric, visibility, and sensor-support indicators identified several categories with consistently unfavorable observation characteristics, including pedestrian-related classes, debris, and pushable objects. These categories exhibit smaller object sizes, lower visibility, and weaker sensor support than most other categories, making them among the most challenging objects to observe in the dataset. Together, they represent approximately 21.3% of all annotations.

Overall, the EDA indicates that object observability is strongly influenced by object size, distance, visibility, and sensor support. Small and partially visible objects are more frequently associated with weak-support conditions, whereas large vehicle classes generally receive stronger camera and LiDAR support. The results further show that camera and LiDAR provide the strongest object-level support across a wide range of observation conditions, motivating their selection for the comparative model analysis in the next chapter.

4 Machine Learning Analysis

4.1 Problem Definition

The objective of this study is to evaluate the impact of camera–LiDAR fusion on 3D object detection performance in the nuScenes dataset.

The task consists of detecting surrounding traffic participants and estimating their 3D bounding boxes, including object class, position, dimensions, and orientation. Two state-of-the-art approaches are compared: CenterPoint [2], a LiDAR-only detector, and BEVFusion [3], a camera–LiDAR fusion model.

The exploratory data analysis showed that camera and LiDAR provide the strongest object-level support across distance ranges, visibility conditions, and object categories, motivating the selection of these two modalities for further investigation.

Due to computational constraints, reduced training subsets containing 20% and 40% of the original training data were created while preserving the official validation set.

Models are evaluated using the official nuScenes detection metrics to determine whether camera–LiDAR fusion provides measurable improvements over a LiDAR-only baseline. Although the nuScenes dataset contains 23 annotated object categories, the official detection benchmark evaluates performance on ten detection classes obtained by grouping several related categories. Accordingly, both models were trained and evaluated on the ten benchmark classes: car, truck, construction vehicle, bus, trailer, barrier, motorcycle, bicycle, pedestrian, and traffic cone.

4.2 Dataset Preparation

The experiments were conducted using the nuScenes v1.0-trainval dataset. Two reduced training subsets containing 20% and 40% of the original training scenes were created to enable a controlled comparison of sensing modalities under limited computational resources. The resulting dataset splits are summarized in Table 4.

The subsets were generated at the scene level to preserve the temporal structure of the driving sequences and maintain complete sensor and annotation information for each selected scene. Scene selection was performed deterministically to ensure reproducibility. The official nuScenes validation set was kept unchanged and used for the evaluation of all models.

Dataset Split	Number of Scenes
Full Training Set	700
20% Training Subset	140
40% Training Subset	280
Validation Set	150

Table 4 - Dataset splits used in the experiments.

4.3 CenterPoint

CenterPoint was selected as the baseline model for this study. It is a state-of-the-art LiDAR-based 3D object detector that has demonstrated strong performance on the nuScenes benchmark.

The model follows a center-based detection approach in which LiDAR point clouds are voxelized and encoded into a Bird's-Eye View (BEV) representation. A convolutional backbone extracts spatial features, and a detection head predicts object centers, classes, dimensions, orientation, and velocity. An overview of the architecture is shown in Figure 21.

The MMDetection3D implementation was used without architectural modifications. The model was trained on the 20% and 40% nuScenes subsets and evaluated using the official nuScenes metrics.

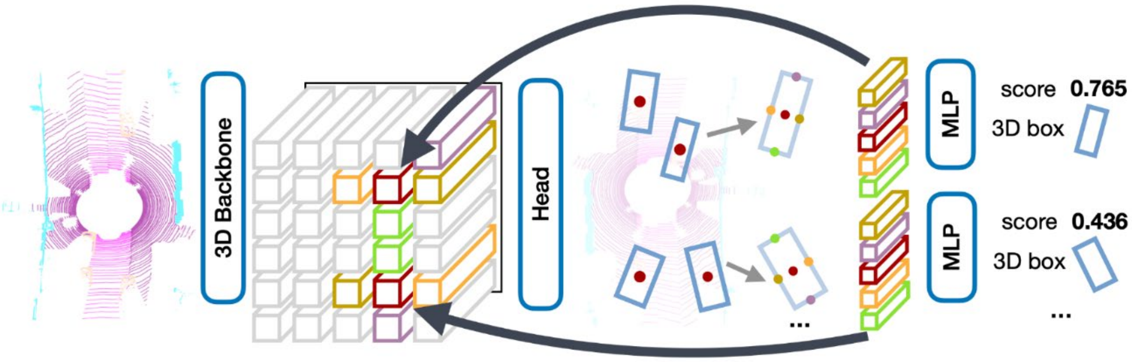

Figure 21 - Overview of the CenterPoint architecture [2]

4.4 BEVFusion

BEVFusion was selected as the multimodal model for this study. It combines camera and LiDAR information within a shared Bird's-Eye View (BEV) representation and achieves state-of-the-art performance on the nuScenes benchmark.

The architecture processes images and LiDAR point clouds through separate feature extraction pipelines before transforming both modalities into a shared BEV representation. The fused BEV features are then passed to a detection head that predicts object classes and 3D bounding boxes. An overview of the architecture is shown in Figure 22.

By combining semantic information from cameras with geometric information from LiDAR, BEVFusion exploits the complementary strengths of both modalities, making it well suited to this study.

The MMDetection3D implementation was used without architectural modifications. The model was trained on the 20% and 40% nuScenes subsets and evaluated using the official nuScenes metrics.

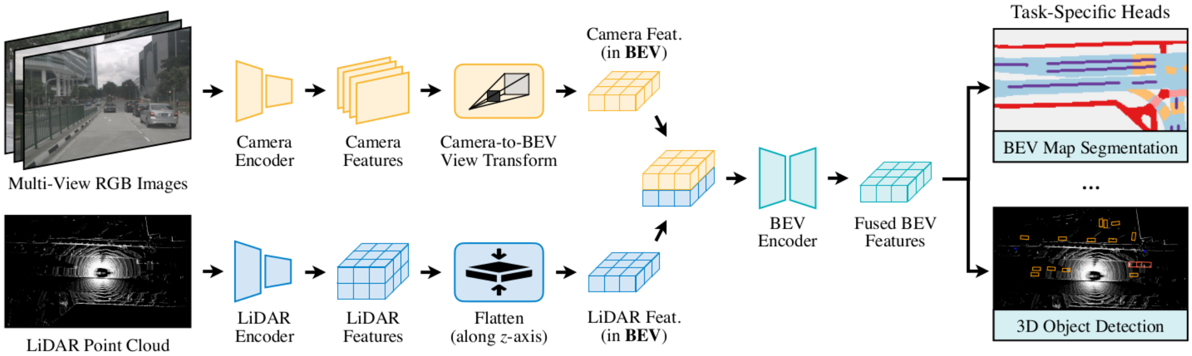

Figure 22 - Overview of the BEVFusion architecture [3].

4.5 Experimental Setup

All experiments were conducted using the MMDetection3D framework from the OpenMMLab ecosystem. Training was performed on the Ubelix HPC cluster using NVIDIA RTX 4090 and NVIDIA A100 GPUs.

Four experiments were conducted to evaluate the effects of sensing modality and training dataset size. The experimental configurations are summarized in Table 5.

Experiment	Model	Training Subset
E1	CenterPoint	20%
E2	CenterPoint	40%
E3	BEVFusion	20%
E4	BEVFusion	40%

Table 5 - Summary of the experiments conducted in this study.

The complete project repository is publicly available on GitHub[3] and contains the data-processing pipeline, exploratory analysis notebooks, experiment configurations, training scripts, and evaluation code used in this study.

[3] https://github.com/darkcyan-ugli-fruit/nuscenes-multimodal-learning

4.6 Training and Evaluation Metrics

Model performance was evaluated using the official nuScenes detection benchmark metrics. The primary evaluation metrics are the nuScenes Detection Score (NDS) and the mean Average Precision (mAP).

The mAP metric measures the detection accuracy across all object classes and is computed using distance-based matching criteria between predicted and ground-truth objects.

The NDS metric provides a more comprehensive assessment by combining mAP with localization, scale, orientation, velocity, and attribute estimation metrics. It is the principal ranking metric used by the nuScenes benchmark.

In addition to NDS and mAP, the benchmark reports five true-positive error metrics:

- mATE: Mean Average Translation Error, measuring localization accuracy.
- mASE: Mean Average Scale Error, measuring object size estimation accuracy.
- mAOE: Mean Average Orientation Error, measuring heading estimation accuracy.
- mAVE: Mean Average Velocity Error, measuring velocity estimation accuracy.
- mAAE: Mean Average Attribute Error, measuring attribute classification accuracy.

5 Results Discussion

5.1 Training Dynamics

Training dynamics were analyzed using the evolution of the nuScenes Detection Score (NDS) and mean Average Precision (mAP) over the 10 training epochs.

Figure 23 shows the progression of NDS during training. All experiments exhibit rapid improvements during the first epochs followed by a gradual convergence. Performance gains become smaller during the later stages of training, indicating that the models are approaching convergence.

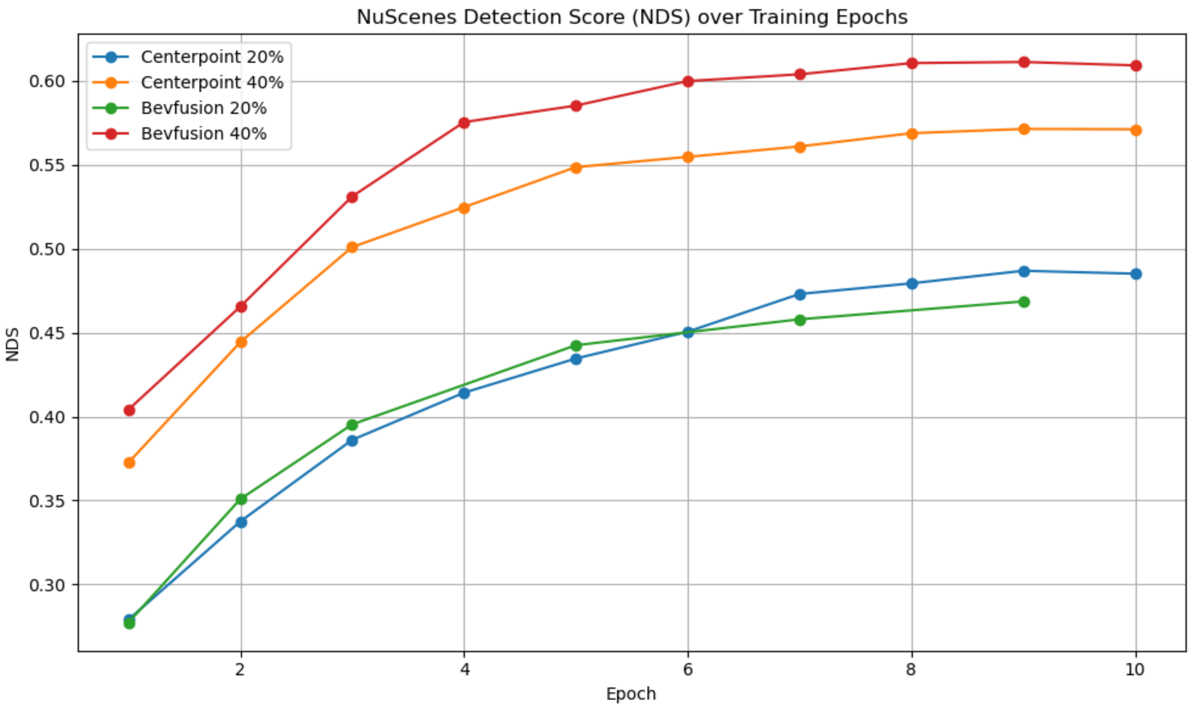

Figure 23 - Evolution of the nuScenes Detection Score (NDS) over training epochs for CenterPoint and BEVFusion trained on the 20% and 40% nuScenes subsets.

A similar trend is observed for mAP in Figure 24. Detection accuracy increases rapidly before reaching a plateau near the end of training. No significant performance instability is observed for any of the experimental configurations.

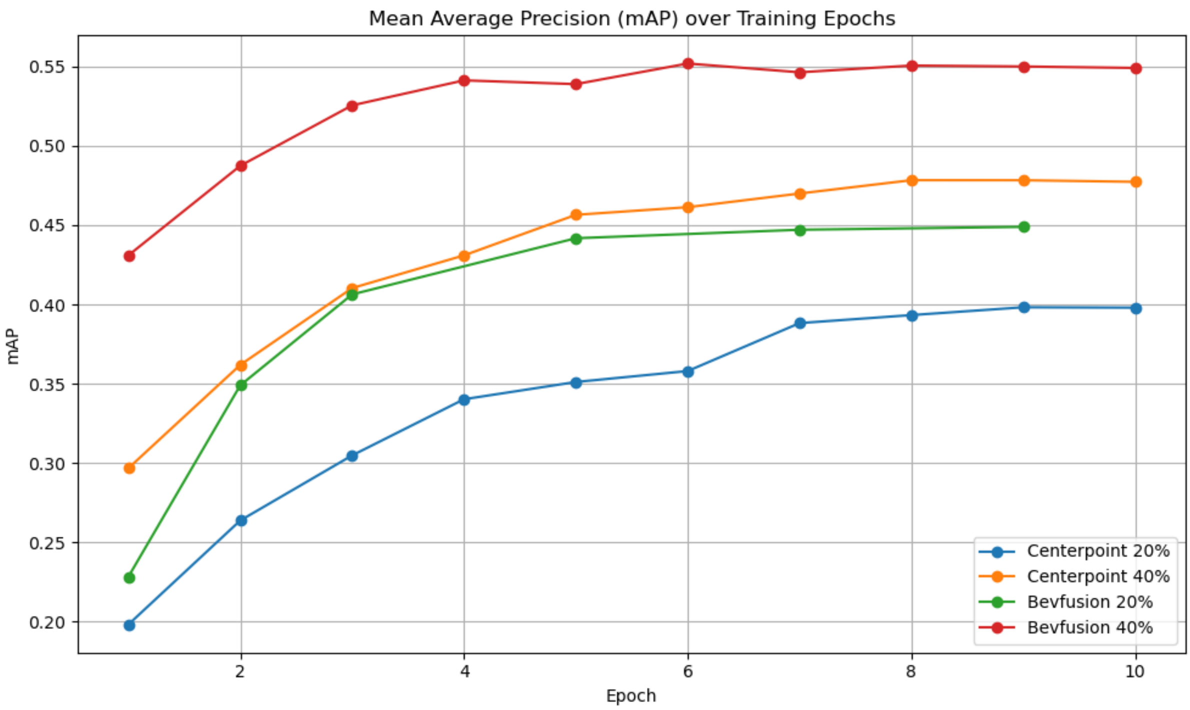

Figure 24 - Evolution of mean Average Precision (mAP) over training epochs for CenterPoint and BEVFusion trained on the 20% and 40% nuScenes subsets.

Overall, all experiments exhibited stable convergence, with performance improvements becoming progressively smaller during the later epochs. The selected 10-epoch training schedule was therefore considered sufficient for a comparative evaluation of the different model and dataset configurations.

5.2 Global Performance

The final performance obtained for the four experimental configurations is summarized in Table 6 and Table 7. These metrics provide an overall comparison of the different models and training subsets.

Table 6 presents the primary nuScenes evaluation metrics, namely the nuScenes Detection Score (NDS) and mean Average Precision (mAP). Performance improved with training dataset size for both models. Among the evaluated configurations, BEVFusion trained on the 40% subset achieved the highest overall performance, obtaining an NDS of 0.609 and an mAP of 0.549, followed by CenterPoint trained on the 40% subset with an NDS of 0.571 and an mAP of 0.477.

Experiment	**NDS ↑**	**mAP ↑**
CenterPoint 20%	0.485	0.398
CenterPoint 40%	0.571	0.477
BEVFusion 20%	0.469	0.449
BEVFusion 40%	**0.609**	**0.549**

Table 6 - Final nuScenes detection performance metrics for all experiments. Higher values indicate better performance.

Table 7 summarizes the nuScenes true-positive error metrics, where lower values indicate more accurate predictions. BEVFusion trained on the 40% subset achieved the lowest translation (mATE), velocity (mAVE), and attribute (mAAE) errors, whereas CenterPoint trained on the 40% subset achieved the lowest scale (mASE) and orientation (mAOE) errors. These results suggest that the performance advantage of BEVFusion is primarily associated with improvements in localization and attribute estimation rather than scale or orientation prediction.

Experiment	mATE ↓	mASE ↓	mAOE ↓	mAVE ↓	mAAE ↓
CenterPoint 20%	0.377	0.2827	0.6237	0.6239	0.2313
CenterPoint 40%	0.3488	**0.2725**	**0.4419**	0.4004	0.2111
BEVFusion 20%	0.37	0.3014	0.6561	0.9719	0.259
BEVFusion 40%	**0.3141**	0.2824	0.4798	**0.3772**	**0.1993**

Table 7 - Final nuScenes true-positive error metrics for all experiments. Lower values indicate better performance.

Overall, BEVFusion trained on the 40% subset delivered the strongest performance across most evaluation metrics, achieving the highest NDS and mAP scores as well as the lowest translation (mATE), velocity (mAVE), and attribute errors (mAAE).

5.3 Per-Class Performance

Table 8 summarizes the average precision (AP) obtained for each object category across the four experimental configurations. Considerable performance differences can be observed between object classes.

Class	BEVFusion 40%	BEVFusion 20%	CenterPoint 40%	CenterPoint 20%
Car	**0.838**	0.803	**0.798**	**0.747**
Truck	0.48	0.335	0.462	0.378
Bus	0.565	0.381	0.615	0.554
Trailer	0.252	0.139	0.247	0.178
Construction Vehicle	**0.19**	**0.107**	**0.099**	**0.099**
Pedestrian	0.838	**0.823**	0.783	0.703
Motorcycle	0.624	0.514	0.44	0.372
Bicycle	0.47	0.282	0.277	0.172
Traffic Cone	0.718	0.654	0.545	0.426
Barrier	0.516	0.451	0.507	0.351

Table 8 - Per-class average precision (AP) for all experiments. Higher values indicate better performance.

The highest detection performance is observed for cars and pedestrians, with AP values exceeding 0.8 for the best-performing configurations. These classes are among the most frequent categories in the dataset and receive strong camera and LiDAR support across a wide range of observation conditions.

In contrast, trailers and construction vehicles remain the most challenging categories, with AP values below 0.3 across all experiments. These classes are relatively infrequent and exhibited substantial variability in object dimensions in the exploratory analysis, which may contribute to their consistently lower detection performance.

The results also suggest that the relative performance of the models varies across object categories. While some classes achieve similar performance across configurations, others exhibit substantial differences, indicating that the benefits of additional training data and multimodal fusion are not distributed uniformly across all categories.

5.4 Impact of Dataset Size

To evaluate the influence of training data quantity, the per-class average precision (AP) obtained with the 40% training subset was compared with the AP obtained with the 20% subset for the same model. The reported dataset gain therefore corresponds to:

Dataset gain = AP(40% subset) − AP(20% subset)

This calculation was performed separately for BEVFusion and CenterPoint. Positive values indicate that increasing the training subset from 20% to 40% improved the detection performance for a given object category. The resulting per-class gains are summarized in Table 9.

Class	BEVFusion Gain	CenterPoint Gain
Bicycle	0.188	0.105
Bus	0.184	0.061
Truck	0.145	0.084
Trailer	0.113	0.069
Motorcycle	0.11	0.068
Construction Vehicle	0.083	0
Barrier	0.065	0.156
Traffic Cone	0.064	0.119
Car	0.035	0.051
Pedestrian	0.015	0.08

Table 9 - Per-class AP improvement obtained when increasing the training subset from 20% to 40%.

For most object categories, the performance gains observed for BEVFusion were larger than those obtained with CenterPoint.

In contrast, cars and pedestrians showed relatively small improvements despite their strong overall performance. This may indicate that these common classes are already well learned with the smaller training subset.

Overall, the results demonstrate that increasing the amount of training data provides consistent performance gains and is particularly beneficial for challenging or less frequent object categories.

5.5 Impact of Camera-LiDAR Fusion

To evaluate the contribution of multimodal fusion, the per-class performance of BEVFusion was compared with the LiDAR-only CenterPoint baseline at the same dataset size. The reported model gap therefore corresponds to:

Model gap = AP(BEVFusion) − AP(CenterPoint)

This difference was computed separately for the 20% and 40% training subsets and summarized in Table 10. Positive values indicate that BEVFusion achieved higher AP than CenterPoint, while negative values indicate an advantage for the LiDAR-only baseline.

Class	20% Gap	40% Gap
Bicycle	0.11	0.193
Motorcycle	0.142	0.184
Traffic Cone	0.228	0.173
Construction Vehicle	0.008	0.091
Pedestrian	0.12	0.055
Car	0.056	0.04
Truck	-0.043	0.018
Barrier	0.1	0.009
Trailer	-0.039	0.005
Bus	-0.173	-0.05

Table 10 - Per-class AP improvement of BEVFusion relative to CenterPoint. Positive values indicate an advantage for BEVFusion.

The largest fusion gains were observed for bicycles, motorcycles, and traffic cones. These categories were identified in the exploratory analysis as receiving comparatively weaker sensor support and are generally smaller than large vehicle classes, suggesting that camera information provides complementary cues not fully captured by LiDAR alone.

This observation is consistent with the dataset-size analysis, which showed larger performance gains for BEVFusion than for CenterPoint when increasing the training subset from 20% to 40%.

Overall, the results confirm that camera–LiDAR fusion provides a measurable benefit for several object classes, particularly small and visually distinctive objects. These findings are consistent with the exploratory data analysis, which highlighted the complementary strengths of camera and LiDAR sensing modalities.

5.6 Limitations and Uncertainty

Several limitations should be considered when interpreting the results. Due to computational constraints, experiments were conducted on 20% and 40% training subsets rather than the full nuScenes dataset. In addition, all models were trained for 10 epochs instead of the standard 20-epoch training schedule, and no hyperparameter optimization was performed. Consequently, the reported results should be interpreted as comparative rather than fully optimized performance estimates.

Each configuration was evaluated using a single training run, preventing an assessment of run-to-run variability. Finally, performance was measured on the nuScenes validation set only, as evaluation on the official test set requires submission to the benchmark server. Consequently, the reported results cannot be directly compared with published benchmark rankings.

6 Conclusion

The objective of this study was to evaluate the impact of camera–LiDAR fusion on 3D object detection performance using the nuScenes dataset.

To investigate this question, the LiDAR-only CenterPoint detector and the camera–LiDAR BEVFusion model were trained and evaluated on reduced nuScenes training subsets containing 20% and 40% of the available training data.

This analysis showed that camera and LiDAR provide the strongest object-level support across a wide range of observation conditions and frequently provide useful observations simultaneously, motivating their selection for the comparative study.

The experimental results showed that increasing the amount of training data consistently improved detection performance for both models. Among the evaluated configurations, BEVFusion trained on the 40% subset achieved the highest overall performance, obtaining the best NDS and mAP scores while also achieving the lowest translation, velocity, and attribute errors.

These findings are consistent with the exploratory data analysis, which showed that smaller object categories generally receive weaker sensor support and therefore benefit more from the complementary semantic and geometric information provided by camera and LiDAR observations.

Overall, the results indicate that camera–LiDAR fusion can provide measurable improvements in 3D object detection performance, particularly for small and weakly supported objects, and when sufficient training data is available.

6.1 Outlook

Several directions for future work can be identified. Training the models on the full nuScenes training set using the standard training schedule would provide a more representative assessment of their performance. Additional experiments could further investigate the impact of hyperparameter optimization and training duration.

From a methodological perspective, the analysis could be extended to other multimodal architectures and fusion strategies. The contribution of individual sensor modalities, including radar, could also be evaluated using model-based rather than dataset-level analyses.

Finally, evaluation on the official nuScenes test benchmark would enable direct comparison with published state-of-the-art methods and provide a more robust assessment of model generalization.

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8 Bibliography

[1] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan and O. Beijbom, "nuscenes: A multimodal dataset for autonomous driving," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020.

[2] T. Yin, X. Zhou and P. Krähenbühl, "Center-based 3D Object Detection and Tracking," *CVPR,* 2021.

[3] Z. Liu, H. Tang, A. Amini, X. Yang, H. Mao, D. Rus and S. Han, "BEVFusion: Multi-Task Multi-Sensor Fusion with Unified Bird's-Eye View Representation," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.